

PROGRESSIVE MULTI-VIEW FUSION FOR 3D HUMAN POSE ESTIMATION

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ABSTRACT

In multi-view 3D human pose estimation (HPE), viewpoint images have large variability due to factors like camera angles and occlusion, making feature extraction and fusion across viewpoints challenging. To address these concerns, we propose a progressive multi-view 3D HPE transformer framework, which achieves effective intra-view pose feature extraction and cross-view fusion by embedding various multi-view fusion methods in the feature extraction process. In order to fully extract spatial semantic features of human joints, we first construct a cross-view spatial fusion module performing spatial feature fusion across adjacent views while mining useful spatial knowledge. To enhance the pose features and alleviate the depth ambiguity problem, we further develop a multi-view spatial-temporal fusion module to extract effective temporal contextual information within the viewpoint and fuse spatial-temporal features across multiple viewpoints. Extensive experiments on two popular 3D HPE benchmarks validate the efficacy and superiority of our method. It outperforms several state-of-the-art methods, effectively alleviates depth ambiguity, and improves 3D pose accuracy without providing camera parameters or complex loss functions.

Index Terms— Multi-view fusion, 3D human pose estimation, transformer

1. INTRODUCTION

Multi-view 3D human pose estimation (HPE) [1, 2, 3, 4] is a hotspot and significant research area in 3D HPE field. It determines the 3D coordinates of major joint points in the human body using multiple images captured by several synchronized cameras. It helps alleviate depth ambiguity of single-view 3D HPE [5, 6, 7] and improves model performance in challenging situations like occlusion and crowding.

Multi-view 3D HPE is currently being studied primarily in terms of multi-view feature fusion [8, 9], multi-view consistency construction [10, 11], and epipolar geometry [12,

13]. Multi-view consistency construction methods are typically based on complex constraints or loss functions, which involve numerous influencing factors and limit the model accordingly. The epipolar geometry methods generally have higher pose prediction accuracy, but they greatly rely on camera parameters of fixed scenes, making it difficult to adapt to different scenes. The multi-view feature fusion method is more adaptable and is the focus of this paper. However, due to complex factors such as camera angles, illumination, and occlusion, there are large discrepancies between images from different viewpoints, which makes it challenging to extract effective information of intra-view features and difficult to perform direct and effective fusion of inter-view features.

To address the above problems, this paper proposes a progressive multi-view fusion framework for 3D HPE. By embedding various multi-view fusion methods in the process of human pose feature extraction, we achieve representative pose feature extraction within viewpoints and effective feature fusion across viewpoints, which significantly improves 3D HPE performance. To explore the deep spatial semantic features, we first model the spatial structure dependencies among human joints for each viewpoint using the multi-head self-attention of the spatial transformer encoder. Then, we build a multi-head cross-view attention mechanism to establish the association between adjacent viewpoint features in order to achieve consistent and distinct spatial feature fusion across viewpoints. On this basis, we employ the temporal transformer encoder to mine the intra-view feature temporal contextual relationships of each viewpoint. These temporal features across viewpoints are further refined and fused with spatial features through a feature refinement module.

The main contributions of this paper are: (1) A progressive transformer-based multi-view 3D HPE framework is built, which effectively mitigates depth ambiguity and improves 3D pose prediction accuracy by multi-level embedding of intra-view feature extraction and cross-view feature fusion. (2) Various cross-view spatial and temporal feature fusion methods are constructed to strengthen inter-view feature correlation and achieve consistent and distinctive feature fusion across multiple viewpoints. (3) The experimental re-

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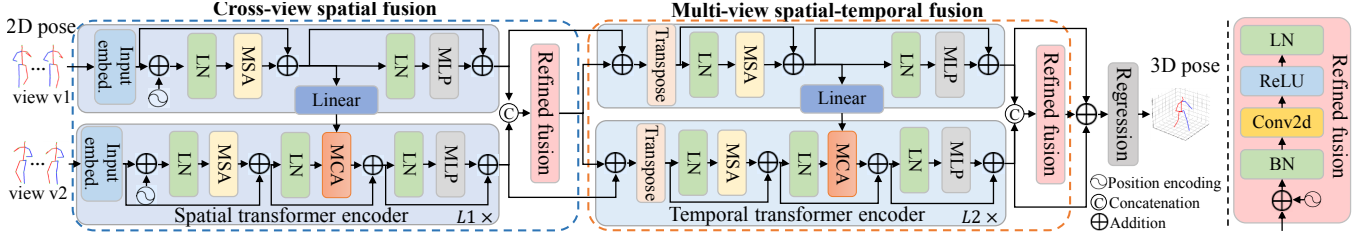


Fig. 1: The architecture of the proposed method. Cross-view spatial fusion module is constructed to achieve intra-view extraction and cross-view fusion of spatial features. Multi-view spatial-temporal fusion module is further built for temporal contextual information supplement within a single viewpoint and spatial-temporal feature fusion among multiple viewpoints.

sults on two popular benchmarks of 3D HPE demonstrate the superior performance of our method beyond many state-of-the-art multi-view 3D HPE methods.

2. METHODOLOGY

For the input sequence containing V views and T frame pose images, we first use an offline 2D pose detector to acquire the 2D pose in each image, and then the proposed method is adopted to obtain the corresponding 3D pose. As shown in Fig 1, our framework is divided into two main parts: Cross-view spatial fusion (CSF) and Multi-view spatial-temporal fusion (MSTF). The CSF is built to extract spatial semantic features within a single viewpoint and model spatial joint dependencies across adjacent viewpoints. The MSTF is created to first mine intra-view temporal contextual relationships of related frames, and then effectively fuse spatial-temporal knowledge across all viewpoints to further enhance the representativeness of final features. The details are as follows.

2.1. Preliminary

Our network is built based on basic components of transformer [14], such as MHA and MLP.

MHA. The input feature $X \in \mathbb{R}^{N \times d}$ is first linearly transformed to generate query Q , key K , and value V . Then the attention matrix among them is calculated through Scaled Dot-Product Attention, which is denoted as:

$$\text{Attention}(Q, K, V) = \text{Softmax} \left(\frac{QK^\top}{\sqrt{d}} \right) V, \quad (1)$$

where N is the input sample number and d is the feature dimension. By projecting Q , K , and V matrices h times using different linear projections, MHA performs attention functions in parallel and concatenates these heads together.

$$\begin{aligned} \text{MHA}(Q, K, V) &= \text{Concat}(H_1, \dots, H_i, \dots, H_h) W_{\text{out}} \\ \text{where } H_i &= \text{Attention}(Q_i, K_i, V_i), i \in [1, \dots, H]. \end{aligned} \quad (2)$$

MLP. It is made up of two linear layers that are used for feature nonlinearization and dimension transformation.

$$\text{MLP}(x) = \sigma(xW_1 + b_1)W_2 + b_2. \quad (3)$$

where σ is the activation function of $GELU$. The weights of the two linear layers are represented by W_1 and W_2 , respectively. The corresponding bias terms are b_1 and b_2 .

2.2. Cross-view spatial fusion

In order to fully exploit intra-view spatial relationships among human body joints and enhance cross-view spatial semantic dependencies, the cross-view spatial fusion (CSF) module is built. It extracts spatial features that characterize joint semantic information within each viewpoint and perform preliminary spatial feature fusion across adjacent viewpoints.

Intra-view spatial feature extraction. The transformer's encoder [14] is adopted to extract pose spatial features within each viewpoint. We add a learnable spatial position matrix E_{SPos} of all joint points to preserve their original spatial structure. The spatial dependencies of these points are modeled using a multi-head self-attention (MSA) mechanism, in which Q , K , and V matrices are the same viewpoint features transformed by different linear functions. Function $\mathcal{F}$ is used for dimension transformation of input feature, and output feature Y_{vi} of viewpoint vi through the process is obtained as:

$$\begin{aligned} X'_{vi} &= \mathcal{F}(X_{vi}) + E_{SPos}, \\ X''_{vi} &= \text{MSA}(\text{LN}(X'_{vi}), \text{LN}(X'_{vi}), \text{LN}(X'_{vi})) + X'_{vi}, \\ Y_{vi} &= \text{MLP}(\text{LN}(X''_{vi})) + X''_{vi}. \end{aligned} \quad (4)$$

Cross-view spatial feature fusion. A multi-head cross-attention (MCA) module is constructed to bridge the spatial feature relationships across adjacent viewpoints, in which the multi-head self-attention features of each viewpoint are used as the inputs. This module is embedded in the intra-view feature extraction process. If we fuse the features of view v_1 into view v_2 via the MCA module, the fusion feature Y_{v2} is:

$$\begin{aligned} X'''_{v2} &= \text{MCA}(F_{c1}(X''_{v1}), F_{c2}(X''_{v1}), \text{LN}(X''_{v2})) + X''_{v2}, \\ Y_{v2} &= \text{MLP}(\text{LN}(X'''_{v2})) + X'''_{v2}. \end{aligned} \quad (5)$$

Where F_{c1} and F_{c2} are distinct linear functions. For further feature optimization, Y_{vi} are input to a refined fusion module with view position encoding E_{VPos} . The final output spatial

feature Y_S after the CSF module is expressed as:

$$\begin{aligned} Y' &= \text{Concat}(Y_{v1}, \dots, Y_{vi}, \dots, Y_V) + E_{VPos}, \\ Y_S &= \text{LN}(\text{ReLU}(\text{Con}(\text{BN}(Y')))). \end{aligned} \quad (6)$$

2.3. Multi-view spatial-temporal fusion

To further enhance the characterization of pose features and alleviate depth ambiguity, we build a multi-view spatial-temporal fusion (MSTF) module. The cross-view spatial fusion features Y_{vi} and Y_S are transposed to temporal channel and added with temporal frame position encoding E_{TPos} . For every view, transformer's encoder is used to model temporal dependencies among related frames in the sequence. Temporal contextual knowledge is extracted to supplement missing pose depth data in the target image. The MCA module is also embedded here for feature interactions and fusion across viewpoints. The output feature of each viewpoint through this process can be achieved as Eq. 4 and Eq.5, respectively. The spatial-temporal fusion features across various viewpoints are further optimized by feature refined fusion module as Eq.6.

The final 3D pose is then got using feature regression with an MLP layer. Through the progressive embedding of intra-view feature extraction and cross-view spatial-temporal feature fusion, representative pose features containing consistent and distinctive knowledge among multiple viewpoints are obtained and used to improve 3D HPE performance.

3. EXPERIMENTS

3.1. Implementations

Human3.6M [19] is the largest and most representative 3D HPE benchmark, containing 3.6 million 3D human pose images and annotations with 17 human joint points. They are captured by four synchronized cameras in an indoor environment, involving 15 actions. Subjects S1, S5, S6, S7, S8 are used for training, S9 and S11 are for testing. **Ski-Pose PTZ-Camera** [20] shows competitive alpine skiers performing giant slalom runs. It contains challenging real-world scenarios images captured by six viewpoint cameras. The model is trained and tested using official training and testing datasets.

Model performance is evaluated using *Protocol1* and *Protocol2*. *Protocol1* (MPJPE) measures the mean Euclidean distance in millimeters between ground-truth and estimated joint points. *Protocol2* (P-MPJPE) measures the Procrustes-MPJPE error in millimeters after alignment.

Experiments are conducted on PyTorch using four GeForce RTX 2080Ti GPUs. The Amsgrad optimizer is used with a weight decay of 0.1. The initial learning rate, batch size, and training epoch are set to 0.0002, 512, and 50, respectively. The learning shrink factor α after each epoch is 0.98. CPN [24] is adopted to detect 2D pose. The number of spatial and temporal transformer encoders is 3 and 2, respectively. The L2 loss is used to train the model.

3.2. Comparison with state-of-the-art

Results on Human3.6M. Table 1 shows our results compared with related multi-view 3D HPE approaches on Human3.6M dataset. It demonstrates that the proposed method achieves state-of-the-art results, with MPJPE reduced by 0.8mm compared to [1]. When ground-truth is used as 2D pose input, MPJPE decreases by 16.1mm compared to Ours(CPN), indicating that 2D pose is important in the 2d-to-3d lifting process and that a better 2D pose detector will aid in 3D HPE performance improvements. It is worth noting that our method does not use any camera parameters, additional training data, or complicated loss function, which distinguishes it from competing approaches. With a simple framework and progressive cross-view feature fusion modules, it effectively alleviates depth ambiguity and improves model performance.

Results on Ski-Pose. Table 2 compares the results of our method to that of related recent approaches on challenging outdoor data. Cases of finetuning on Human3.6M and testing on Ski-Pose are investigated, as are both training and testing on Ski-Pose. In both cases, our method surpasses all the compared jobs. Our MPJPE improves by 23.1mm compared to [18] in the first case, noting that more categories of training data help to improve model performance. In the second case, our method is also very competitive, increasing MPJPE by 14.0mm more than [18]. Visual results are shown in Fig. 3.

3.3. Ablation study

Impact of fusion modules. Table 3 compares the effects of various fusion modules on 3D HPE performance, including spatial and temporal transformer encoder (STE and TTE), RF, and MCA in CSF and MSTF. The results show that removing either the CSF or the MSTF significantly degrades model performance. STE has a greater impact on the model than TTE, and RF in CSF contributes more than it does in MSTF, implying that valid spatial semantic information characterizes pose features better than temporal data. Model performance suffers in the absence of MCA, demonstrating that the interaction of information from different viewpoints aids in mining more effective features and improving 3D pose estimation accuracy.

Impact of views. Table 4 depicts the effects of varying input viewpoint numbers on model performance. The model's MPJPE steadily decreases as the viewpoint increases. It implies that multi-view data can help to supplement missing depth knowledge and improve model performance. Furthermore, it illustrates that our framework can receive any number of viewpoints without being constrained by a fixed number of views, making it broadly applicable for multi-view fusion.

Impact of temporal receptive fields (TRFs). Fig. 2 shows model performance improves as TRF increases, implying that related temporal information facilitates the model. When TRF=1, our model performs well, indicating that our method can obtain fairly rich pose semantic features even when helpful temporal data is unavailable. As TRF grows,

Methods	Dir.	Disc.	Eat	Greet	Phone	Photo	Pose	Purch	Sit	SitD.	Smoke	Wait	WalkD.	Walk	WalkT.	Avg.
Tome [15]	43.3	49.6	42.0	48.8	51.1	64.3	40.3	43.3	66.0	95.2	50.2	52.2	51.1	43.9	45.3	52.8
Kadkhodamohammadi [16]	39.4	46.9	41.0	42.7	53.6	54.8	41.4	50.0	59.9	78.8	49.8	46.2	51.1	40.5	41.0	49.1
Luvizon [17]	31.0	33.0	41.0	34.0	41.0	37.0	37.0	51.0	56.0	43.0	44.0	37.0	33.0	42.0	32.0	39.0
Gorden [18]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	31.7
Qiu [4]	28.9	32.5	26.6	28.1	28.3	29.3	28.0	36.8	42.0	30.5	35.6	30.0	28.3	30.0	30.5	31.2
Remelli [3]	27.3	32.1	25.0	26.5	29.3	35.4	28.8	31.6	36.4	31.7	31.2	29.9	26.9	33.7	30.4	30.2
Bultmann [2]	27.1	29.9	27.0	26.5	31.3	28.9	27.1	29.8	36.5	36.0	30.8	29.3	29.7	27.3	26.3	29.8
Bartol [1]	27.5	28.4	29.3	27.5	30.1	28.1	27.9	30.8	32.9	32.5	30.8	29.4	28.5	30.5	30.1	29.1
Ours (CPN,T=27)	26.1	28.2	24.5	26.9	28.7	31.2	26.4	28.2	31.5	33.6	29.4	25.8	28.8	27.4	27.0	28.3
Ours (GT,T=27)	12.3	12.8	11.2	12.8	11.5	13.5	12.5	12.7	11.9	11.5	12.1	12.1	12.6	11.7	12.1	12.2

Table 1: Comparisons with state-of-the-art 3D HPE methods on Human3.6M with MPJPE(mm). Best in **bold**.

Methods	Training	MPJPE(mm)	P-MPJPE(mm)
Chen [21]	H36M	130.2	108.7
Wandt [22]	Ski	128.1	89.6
Chen [21]	H36M+Ski	99.4	74.7
Rhodin [23]	Ski	85.0	-
Gorden [18]	Ski	65.5	-
Ours	Ski	51.5	40.4
	H36M+Ski	42.4	26.4

Table 2: Results on Ski-Pose dataset. Best in **bold**.

CSF	MSTF	MCA	MPJPE(mm)	P-MPJPE(mm)
STE	RF	TTE	RF	
×	✓	✓	✓	✓
×	×	✓	✓	✓
✓	✓	×	✓	✓
✓	✓	×	×	✓
×	×	✓	✓	✓
✓	✓	✓	✓	×
✓	✓	✓	✓	✓
			28.3	21.6

Table 3: Impacts of model components. Best in **bold**.

view number	MPJPE(mm)	P-MPJPE(mm)
2	31.4	25.0
3	30.0	23.8
4	26.8	21.1

Table 4: Impacts of different viewpoints. Best in **bold**.

FLOPs gradually increase, but even at TRF=81, it is still not large. While model parameters almost do not increase, showing that our method is not sensitive to TRF and has no burden processing multiple frame data.

4. CONCLUSION

In this paper, we proposed a progressive multi-view 3D HPE transformer structure, which effectively improves 3D HPE performance by incorporating various cross-view feature fusion modules into the pose feature extraction process. First, we developed a cross-view spatial fusion module capable of effective intra-view spatial semantic feature extraction and

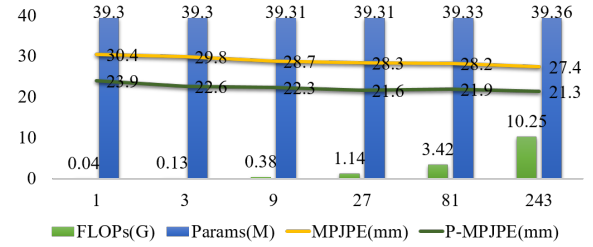


Fig. 2: Impact of temporal receptive fields (TRFs).

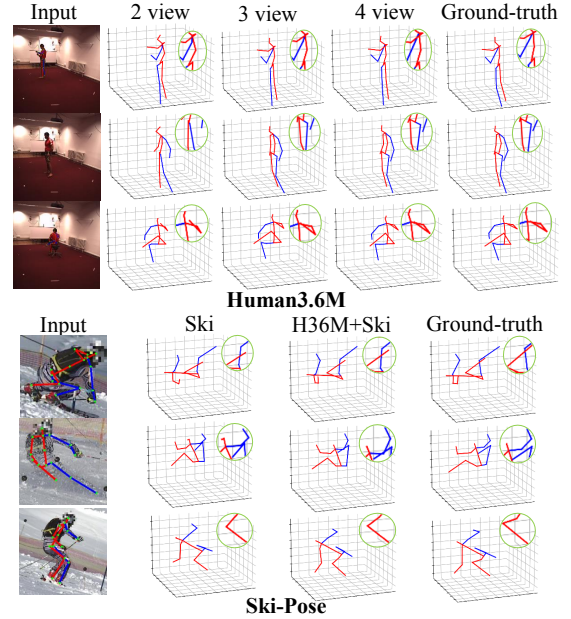


Fig. 3: Visualization results on Human3.6M and Ski-Pose.

adjacent viewpoint fusion. Then we constructed a multi-view spatial-temporal fusion module that successfully extracted and fused sequence temporal information across multiple viewpoints, allowing us to improve pose features and supplement depth information. Numerous experiments showed that our method relied on a simple structure without camera parameters or complex loss functions to effectively reduce depth ambiguity and improve 3D pose prediction accuracy.

5. REFERENCES

- [1] Kristijan Bartol, David Bojanić, Tomislav Petković, and Tomislav Pribanić, “Generalizable human pose triangulation,” in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2022, pp. 11028–11037.
- [2] Simon Bultmann and Sven Behnke, “Real-time multi-view 3d human pose estimation using semantic feedback to smart edge sensors,” in *Robotics: Science and Systems*, 2021.
- [3] Edoardo Remelli, Shangchen Han, Sina Honari, Pascal Fua, and Robert Wang, “Lightweight multi-view 3d pose estimation through camera-disentangled representation,” in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2020.
- [4] Haibo Qiu, Chunyu Wang, Jingdong Wang, Naiyan Wang, and Wenjun Zeng, “Cross view fusion for 3d human pose estimation,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 4342–4351.
- [5] Wenhao Li, Hong Liu, Hao Tang, Pichao Wang, and Luc Van Gool, “Mhformer: Multi-hypothesis transformer for 3d human pose estimation,” in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2022, pp. 13147–13156.
- [6] Lijun Zhang, Xiaohu Shao, Zhenghao Li, Xiang-Dong Zhou, and Yu Shi, “Spatio-temporal attention graph for monocular 3d human pose estimation,” in *2022 IEEE International Conference on Image Processing*, 2022, pp. 1231–1235.
- [7] Ce Zheng, Sijie Zhu, Matias Mendieta, Taojiannan Yang, Chen Chen, and Zhengming Ding, “3d human pose estimation with spatial and temporal transformers,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 11656–11665.
- [8] Rahul Mitra, Nitesh B. Gundavarapu, Abhishek Sharma, and Arjun Jain, “Multiview-consistent semi-supervised learning for 3d human pose estimation,” in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2020, pp. 6907–6916.
- [9] Hui Shuai, Lele Wu, and Qingshan Liu, “Adaptive multi-view and temporal fusing transformer for 3d human pose estimation,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 14, no. 8, pp. 1–14, 2022.
- [10] Inho Chang, Min-Gyu Park, Jaewoo Kim, and Ju Hong Yoon, “Multi-view 3d human pose estimation with self-supervised learning,” in *2021 International Conference on Artificial Intelligence in Information and Communication*, 2021.
- [11] Arij Bouazizi, Julian Wiederer, Ulrich Kressel, and Vasileios Belagiannis, “Self-supervised 3d human pose estimation with multiple-view geometry,” in *2021 16th IEEE International Conference on Automatic Face and Gesture Recognition*, 2021.
- [12] Yihui He, Rui Yan, Katerina Fragkiadaki, and Shoubo I. Yu, “Epipolar transformers,” in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2020, pp. 7779–7788.
- [13] Haoyu Ma, Liangjian Chen, Deying Kong, Zhe Wang, Xingwei Liu, Hao Tang, Xiangyi Yan, Yusheng Xie, Shih-Yao Lin, and Xiaohui Xie, “Transfusion: Cross-view fusion with transformer for 3d human pose estimation,” in *British Machine Vision Conference*, 2021.
- [14] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems 30*, 2017.
- [15] Denis Tome, Matteo Toso, Lourdes Agapito, and Chris Russell, “Rethinking pose in 3d: Multi-stage refinement and recovery for markerless motion capture,” in *2018 International Conference on 3D Vision*, 2018, pp. 474–483.
- [16] Abdolrahim Kadkhodamohammadi and Nicolas Padoy, “A generalizable approach for multi-view 3d human pose regression,” *Machine Vision and Applications*, vol. 32, no. 6, 2021.
- [17] Diogo C. Luvizon, David Picard, and Hedi Tabia, “Consensus-based optimization for 3d human pose estimation in camera coordinates,” *International Journal of Computer Vision*, vol. 130, pp. 869–882, 2022.
- [18] Brian Gordon, Sigal Raab, Guy Azov, Raja Giryes, and Daniel Cohen-Or, “Flex: Parameter-free multi-view 3d human motion reconstruction,” in *European Conference on Computer Vision*, 2022, pp. 176–196.
- [19] Catalin Ionescu, Dragos Papava, Vlad Olaru, and Cristian Sminchisescu, “Human3.6m: Large scale datasets and predictive methods for 3d human sensing in natural environments,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 36, no. 7, 2013.
- [20] Benedikt Fasel, Jörg Spörri, Matthias Gilgien, Geo Boffi, Julien Chardonens, Erich Müller, and Kamiar Aminian, “Three-dimensional body and centre of mass kinematics in alpine ski racing using differential gnss and inertial sensors,” *Remote Sensing*, vol. 8, no. 8, pp. 617, 2016.
- [21] Xipeng Chen, Pengxu Wei, and Liang Lin, “Deductive learning for weakly-supervised 3d human pose estimation via uncalibrated cameras,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2021, pp. 1089–1096.
- [22] Bastian Wandt, Marco Rudolph, Petrisa Zell, Helge Rhodin, and Bodo Rosenhahn, “Canonpose: Self-supervised monocular 3d human pose estimation in the wild,” in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2021, pp. 13294–13304.
- [23] Helge Rhodin, Frédéric Meyer, Jörg Spörri, Erich Müller, Victor Constantin, Pascal Fua, Isinsu Katircioglu, and Mathieu Salzmann, “Learning monocular 3d human pose estimation from multi-view images,” in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2018, pp. 8437–8446.
- [24] Yilun Chen, Zhicheng Wang, Yuxiang Peng, Zhiqiang Zhang, Gang Yu, and Jian Sun, “Cascaded pyramid network for multi-person pose estimation,” in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2018, pp. 7103–7112.